

IOWA LIQUOR SALES COSTO WHOLESALE DATA CLEANING

A DATA CLEANING PROJECT USING POWER QUERY

A	B	C	D	E	F	G	H	I	J	K	L
Invoice Id	Store No	Date Ordered	Store Name	Store Postal Code	Store Address	Store City	Store Zip Code	county_fips_code	County Name	Category Code	Category Name
RINV-059268	4677	9/25/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIGHT BOURBON WHISKIES
RINV-059296	3814	9/26/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIGHT BOURBON WHISKIES
RINV-059297	5666	9/26/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1011300	SINGLE BARREL BOURBON WHIS
INV-870666	10074	9/29/2025	COSTCO WHOLESALE	4000	NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1011200	STRAIGHT BOURBON WHISKIES
INV-870654	4677	9/29/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIGHT BOURBON WHISKIES
INV-871116	3814	9/30/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIGHT BOURBON WHISKIES
INV-871117	3814	9/30/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIGHT BOURBON WHISKIES
INV-871320	5666	9/30/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1011200	STRAIGHT BOURBON WHISKIES
INV-872726	10074	10/6/2025	COSTCO WHOLESALE	4000	NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1011200	STRAIGHT BOURBON WHISKIES
INV-872987	4677	10/6/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011100	BLENDED WHISKIES
INV-872986	4677	10/6/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1012200	SCOTCH WHISKIES
INV-872985	4677	10/6/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1081100	COFFEE LIQUEURS
INV-872988	4677	10/6/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1081200	CREAM LIQUEURS
RINV-059397	10074	10/7/2025	COSTCO WHOLESALE	4000	NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1031100	AMERICAN VODKAS
INV-873202	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIGHT BOURBON WHISKIES
INV-873200	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIGHT BOURBON WHISKIES
INV-873199	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1012200	SCOTCH WHISKIES
INV-873201	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1032100	IMPORTED VODKAS
INV-873198	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1071000	COCKTAILS/RTD
INV-873197	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1081100	COFFEE LIQUEURS
INV-873283	5666	10/7/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1011100	BLENDED WHISKIES
INV-873284	5666	10/7/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1011200	STRAIGHT BOURBON WHISKIES
INV-875214	10074	10/13/2025	COSTCO WHOLESALE	4000	NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1012100	CANADIAN WHISKIES
INV-875509	4677	10/13/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIGHT BOURBON WHISKIES
INV-875510	4677	10/13/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1012400	IRISH WHISKIES
INV-875788	3814	10/14/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1012100	CANADIAN WHISKIES
INV-875871	5666	10/14/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1012200	SCOTCH WHISKIES

Preparation By: Janice Kipkiror

Tools Used: Excel

PROJECT OVERVIEW

This project aims to clean messy data using Power Query. Power Query is a feature in Excel that helps in loading and transforming data. This saves a lot of time in data preparation because the user can load a chunk of data and clean it at a faster rate.

Power Query has its advantages, which are:

- Automation - Whenever you load new data, the steps taken to clean the previous dataset will be replicated.
- Simple and time-saving - Compared to cleaning with complex formulas, Power Query is about clicks, and you are done.

OBJECTIVES

- Cleaning data using Power Query by:
 1. Removing duplicates
 2. Handling missing values
 3. Fixing data types
 4. Removing errors
 5. Remove blank rows

DATASET

This is a secondary dataset from data.gov. It contains 41469 records with 21 features.

TOOLS

- Excel - Power Query

DATA CLEANING

Because this is a data-cleaning project, the following steps were taken with this data.

1. Removal of duplicates - This is especially the Invoice ID, which is the unique field.
2. Removed Blanks
3. Split Columns - Some columns, such as store name and store address, were split.
4. Fixed Data - There are some of the numeric values in sales that were negative, which needed to be made absolute for future operations.
5. Cleaned and trimmed text data types.
6. Reordered columns.
7. Renamed the columns to appropriate names.
8. Loaded and transformed the data into a table.

File

Home

Transform

Add Column

View

Close & Load

Refresh Preview

Advanced Editor

Manage

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Split Column

Group By

Data Type: Text

Use First Row as Headers

Replace Values

Merge Queries

Append Queries

Combine Files

Manage Parameters

Data source settings

New Source

Recent Sources

Query

Transform

Combine

Parameters

Data Sources

New Query

Table.TransformColumnTypes(#{Promoted Headers},{{"invoice_id", type text}, {"ordered_on", type date}, {"store_no", Int64.Type}, {"store_name", type text}, {"store_address", type text},

	Invoice_id	ordered_on	store_no	store_name	store_address	store_city	store_zip_code	county_fips_code	county_name	category_code	category
1	RINV-059268	9/25/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIK
2	RINV-059296	9/26/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIK
3	RINV-059297	9/26/2025	5666	COSTCO WHOLESALE #1325 / DAVENPORT	2790 E 53RD ST	DAVENPORT	52807	19163	SCOTT	1011300	SINGLI
4	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1011200	STRAIK
5	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1011300	SINGLI
6	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1011500	BOTTL
7	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1012100	CANAI
8	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1012300	SINGLI
9	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1022200	100%
10	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1082100	IMPOF
11	INV-870666	9/29/2025	10074	COSTCO WHOLESALE #1630 / ANKENY	4000 NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1701100	TEMPC
12	INV-870654	9/29/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIK
13	INV-870654	9/29/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIK
14	INV-870654	9/29/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIK
15	INV-870654	9/29/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011500	BOTTL
16	INV-870654	9/29/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1012100	CANAI
17	INV-870654	9/29/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1012300	SINGLI
18	INV-870654	9/29/2025	4677	COSTCO WHOLESALE #1111 / CORALVILLE	2900 HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1022200	100%
19	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIK
20	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIK
21	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIK
22	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIK
23	INV-871117	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIK
24	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1012100	CANAI
25	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1012200	SCOTC
26	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1012300	SINGLI
27	INV-871116	9/30/2025	3814	COSTCO WHOLESALE #788 / WDM	7205 MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1022200	100%
28											

2790											
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INV-872986	4677	10/6/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1012200	SCOTCH WHISKIES
INV-872985	4677	10/6/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1081100	COFFEE LIQUEURS
INV-872988	4677	10/6/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1081200	CREAM LIQUEURS
RINV-059397	10074	10/7/2025	COSTCO WHOLESALE	4000	NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1031100	AMERICAN VODKAS
INV-873202	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1011200	STRAIGHT BOURBON WHISKIES
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INV-873197	3814	10/7/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1081100	COFFEE LIQUEURS
INV-873283	5666	10/7/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1011100	BLENDED WHISKIES
INV-873284	5666	10/7/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1011200	STRAIGHT BOURBON WHISKIES
INV-875214	10074	10/13/2025	COSTCO WHOLESALE	4000	NE SPECTRUM DRIVE	ANKENY	50021	19153	POLK	1012100	CANADIAN WHISKIES
INV-875509	4677	10/13/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1011200	STRAIGHT BOURBON WHISKIES
INV-875510	4677	10/13/2025	COSTCO WHOLESALE	2900	HEARTLAND DR	CORALVILLE	52241	19103	JOHNSON	1012400	IRISH WHISKIES
INV-875788	3814	10/14/2025	COSTCO WHOLESALE	7205	MILLS CIVIC PKWY	WEST DES MOINES	50266	19049	DALLAS	1012100	CANADIAN WHISKIES
INV-875871	5666	10/14/2025	COSTCO WHOLESALE	2790	E 53RD ST	DAVENPORT	52807	19163	SCOTT	1012200	SCOTCH WHISKIES
Iowa Liquor Sales Costco Wholes											

KEY INSIGHTS

- Power Query is a time-saving feature, especially when cleaning data.

CHALLENGES

- The data had many columns to split.

CONCLUSION

- The project demonstrated the advantages of using Power Query to prepare your data.

REFERENCES

Data Source

https://catalog.data.gov/dataset/warehouse-and-retail-sales?from_hint=eyJxIjojU2FsZXMiLCJzb3J0IjoicmVsZXZhbmNlIn0%3D
State of Iowa - Iowa Liquor Sales, Costco Wholesale

Github Repository

https://github.com/Janicekipkiror/Warehouse_and_Retail_Sales_Analysis/tree/main
GitHub - Janicekipkiror/Data_Cleaning_Project_With_Power_Query-Excel: This project focuses on cleaning and analyzing warehouse and retail sales data using Power Query and Excel.